doi: 10.1145/3800965.3834287

DESIGNING WITH IMPAIRED TECHNOLOGIES

A GAME JAM OF CONSTRAINTS AND AFFORDANCE SHRINKAGE

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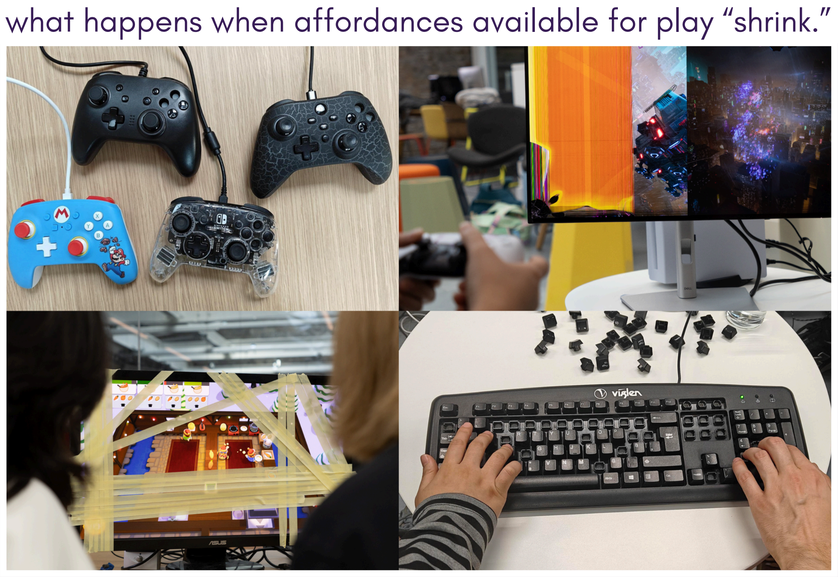
When the sensory interface for play is impaired, how do players adapt their behaviours and strategies? We draw a parallel between the challenges presented by these constrained experiences and the common but too rarely discussed experience of facing barriers to play, whether these barriers come from disabilities, economics or various other factors.

RQ1: What **adaptive strategies** do people use when playing games with impaired technologies?

RQ2: What **types of affordances** emerge when people design games inspired by impaired technologies?

"IMPAIRED TECHNOLOGIES"

These are damaged peripherals that might typically be discarded as e-waste like cracked screens and missing buttons (see below) as **design provocations** to explore what happens when affordances available for play "shrink."



WHAT HAPPENED?

We gathered impaired technologies and hosted a public two-day 9:00 AM - 5:00 PM weekend game jam for adult participants (*n*=9).

DAY 1



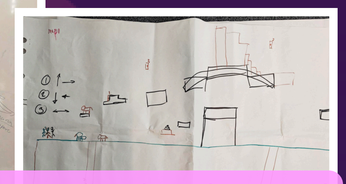
Participants played existing games with impaired technologies

Researchers filmed, observed, and interviewed participants



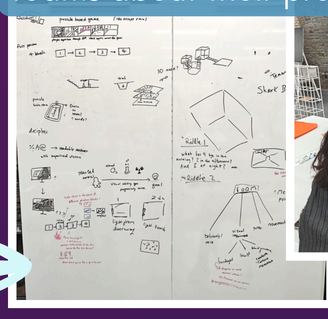
Participants formed three teams of three





The teams brainstormed and started developing ideas for their own games

Researchers filmed and interviewed teams about their process





DAY 2



Participants continued working on their game prototypes within teams

Researchers interviewed, filmed and observed teams

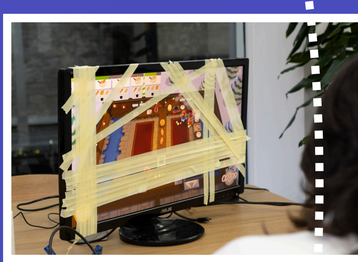


The teams presented their game prototypes to a panel of judges

Researchers asked final questions

FINDINGS: ADAPTIVE PLAY STRATEGIES





In order to play games, participants:

- Changed **game settings**,
- Explored what could work via **trial and error**,
- Manipulated their **bodies** (e.g., leaning, gesturing) or **external tools** (e.g., a pen),
- Found **new forms of fun** by changing game objectives (e.g., firefighting in a kitchen),
- And **collaborated with other players** in order to succeed...

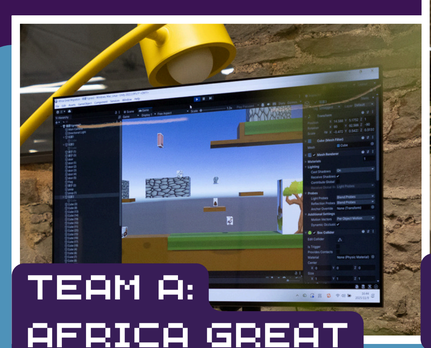
...but in an exception, a participant just **gave up** on playing the game.

FINDINGS: AFFORDANCES THROUGHOUT DESIGN

The three teams created distinctly different games inspired by their experiences playing various existing games with impaired technologies.

Overall, participants:

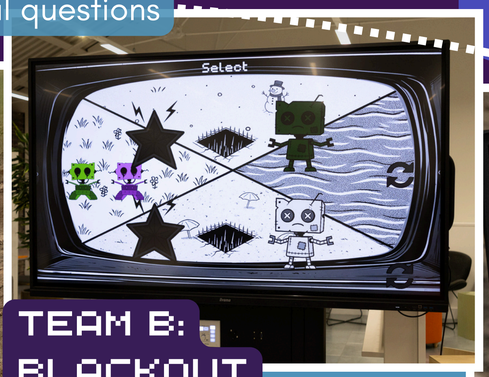
- described **impairments as "bugs,"** flaws, or constraints that were part of **gameplay features**
- **implemented impairment** directly into their game
- tended to **prioritize fun over challenge**



TEAM A: AFRICA GREAT MIGRATION

- 4-player 2D platformer
- Each player only has two directional controls (up/down or left/right)
- Players race to the finish line

Simple, limited and asymmetric player controls



TEAM B: BLACKOUT

- 4-player competitive game
- Players move across four zones to attack each other during blackouts, guided by sounds and visual effects
- Results are only revealed when lights return

Shared uncertainty and inference of feedback



TEAM C: ESCAPE TOMB

- Single-player point-and-click
- Explored visual occlusion as as a central method for solving puzzles
- Players aim to escape the tomb

Hiding or revealing on-screen elements for solving in-game puzzles

DISCUSSION AND FUTURE WORK

- Design provocation encouraged participants to focus on impairment as part of play experience
- Games showcased different interpretations of "impairment"
- Speculative design research method applicable towards a variety of participants and supporting education

DOES OUR WORK INTEREST YOU?

Join our ***Playing with Constraints*** workshop here at CHI PLAY 2026!

This research is funded by a **European Research Council (ERC)** Consolidator fellowship and **UK Research and Innovation (UKRI)** grant no.: EP/Y023676/1 as part of the *inclusiveXplay* project, a UKRI Engineering and Physical Sciences Research Council Doctoral Training Partnership scholarship, and a **Social Sciences and Humanities Research Council (SSHRC)** Canada Graduate Research Scholarship - Doctoral (CGRS-D).